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Space-Object Active Control Mode Inference Using Light Curve Inversion

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Several candidate methods for classifying an agile space-object active control mode are tested using simulated light curve data and a Rao–Blackwellized particle filter. The first measure to discriminate the space-object control mode is a measurement dissimilarity metric, defined as the time integral of the error between the estimated target space-object sensor boresight and the line-of-sight vector to each hypothesized subject. The second measure quantifies the "pointing quality" using the multivariate Gaussian mixture model analog to the Mahalanobis distance, which is computed using the estimated multivariate body angular velocity distributions. It is shown how additional information from the space-object shape model can be combined with radiometric first principles to establish a tracking error threshold between the target space object and the hypothesized subject. Finally, the body angular velocity estimates are used to compute the mass-specific rotational angular momentum and mass-specific rotational kinetic-energy analogs. These analogs are coupled with statistical inference techniques to classify the active control modes of agile space objects.

I. Introduction

IMPROVEMENTS in space domain awareness (SDA) are identified in multiple national policy documents as a top priority to protect the United States and its allies, as well as maintain its economic and diplomatic objectives [1]. The high-level activities of SDA include the detection, tracking, characterization, and analysis of space objects (SOs), as defined in Joint Publication 3-14, titled "Space Operations" [2]. The goal of these activities is to provide timely, actionable evidence of specific space threats and hazards to decision-making processes that protect space services and capabilities.

Although the behavior or intent of a SO cannot be directly measured, the SO shape [3], materials [4], and attitude [5] can be directly estimated. When coupled with subject matter expertise or statistical inference techniques, these quantities could indicate SO behavior or intent. For an SO in low Earth orbit, a shape and attitude estimation is performed extensively using radar-based methods pioneered in the early 1980s [6]. The shape and attitude of a large SO can also be estimated from resolved imagery taken by ground-based optical sensors. However, when SOs are too distant to be characterized by radar facilities or too small to be adequately resolved by ground-based optical sensors, the only data currently available are unresolved imagery [3]. To empower SDA for these important classes of SOs, the contributions of this work focus on inferring a SO's current, active control mode using light curve inversion. This work defines the active control mode as "the classification of a time history of attitude and angular rate states necessary for an SO to track a specific subject." Although this encompasses traditional "operational modes" such as "sun pointing," it also describes more specific behavior such as "tracking SO catalog number 16111," for example.

The earliest work in the public literature on SO operational mode classification, completed by the Lincoln Laboratory, Massachusetts Institute of Technology, uses light curves gathered using the space-based visible sensor to classify a geostationary orbit (GEO) SO in the following categories: nominal in slot, anomalous in slot, nominal moving, anomalous moving, drifting, and graveyard. These classifications were made using Bayesian networks, but little detail is provided due to the operational nature of the work [7].

Another approach outlined in the literature also uses Bayesian networks, where the possible classifications are nominal and anomalous. This work focused on constellations of the SO in the same GEO slot using the classifications to help ameliorate the problem of "cross tagging." Cross tagging occurs when a SO is erroneously identified as another: an occurrence much more likely to happen among a cluster of SOs in GEO sharing the same slot [8]. This work has been extended to multisatellite scenarios where hypothesis testing and sliding window techniques, among others, are proposed to classify SOs using only nightly comparisons of light curves [9].

Despite the modicum of prior work on SO operational mode classification, many similar concepts have been developed in the fault-detection and classification literature. Many of those techniques have already been applied to different SDA-specific problems. Surveying the literature for fault detection as applied to adaptive control, fault control, and systems identification reveals that all change detection algorithms fall into one of three general categories [10].

The first category applies statistical tests to the residuals generated by a Kalman filter. Tests for whiteness, zero mean, and a specified variance all indicate that the system model is representative of the true, physical system. If the residuals fail these tests, and the model accurately reflects the true system, this indicates that the behavior of the physical system has changed [11]. The second category implements two filters in parallel, where statistical tests are applied to the distributions at two discrete time steps. Also referred to as "sliding window" techniques, these tests are applied to certain parameters, including the generalized likelihood ratio, the divergence test, or changes in spectral distance [12]. The third and final category is the multiple-model method, where all hypothesized change times are defined and compared to the filtered estimates. Similar to "matched filtering" techniques, the hypothesis that provides the smallest residuals is taken to be the correct hypothesis [13].

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Perhaps the earliest application of multiple hypothesis methodologies in the SDA field is the multiple-model adaptive estimation technique proposed to determine the most likely shape model of a SO. The core technique, where a bank of filters each operating under a different hypothesis to determine the most likely physical system, is similar to those ideas proposed in the fault-detection literature. Indeed, the most likely physical shape is indicated by the hypothesis that produces the highest likelihood [14].

In a manner similar to multiple-model methods, matched filtering techniques have also been applied to SDA. First applied to increasing the signal-to-noise ratio (SNR) of radar returns, the core concept of a matched filter is that multiple signal "templates" are hypothesized where the correct hypothesis yields the greatest SNR [15]. One corresponding SDA application uses multiple hypothesized templates to detect an unresolved SO in low SNR imagery obtained with optical telescopes [16]. This is particularly useful when numerical propagation errors or sensor pointing uncertainties cause differences between the SO orbit and telescope pointing.

The concept of "active control modes" is motivated by the following scenario: an agile SO under surveillance by an SDA stakeholder, termed the "target" SO, is hypothesized to be actively tracking another SO, called the "subject" SO, with its optical payload. This work emphasizes that an agile SO is one that is maneuvering to produce rotational motion about its center of mass, and not one that is maneuvering to change its orbit plane. The overall goal of the methodologies presented in this work is to identify, from a mutually exclusive but not exhaustive list of hypotheses, which subject is under observation by the agile SO. The methodologies use both estimated attitude states and estimated body angular velocities, such that any sensor that provides these estimates could exploit the contributions enumerated in this work. However, the simulated results assume that a terrestrial telescope produces unresolved images of the target SO. To the authors' knowledge, estimating both the attitude and velocity states of an agile target SO using unresolved imagery requires innovations in light curve inversion previously developed by Coder et al. [17]. Greater detail on the exact filter used to produce the simulated results is given in that manuscript, whereas the current work details how the estimated state distributions could be used to determine the SO active control mode [17].

In solving this scenario, the authors recognize that the knowledge of SDA stakeholders will always contain uncertainty, and they argue that new information should be used to inductively update their beliefs by eliminating incorrect hypotheses. In the fields of computer science and artificial intelligence, the phrases "closed world" and "open world" are used to describe an underlying assumption about the logic used for knowledge representation [18]. In a closed-world system, all knowledge is completely represented; therefore, any statement that cannot be proven true must be false. Conversely, an open-world system acknowledges that uncertainty exists, and any statement that cannot be proven true is either false or unknown [19]. Acknowledging the inherent uncertainty of SDA, the authors make the "open-world assumption" for representing knowledge and present all possible observed subjects as a mutually exclusive and nonexhaustive set of hypotheses. As a result, if the set of hypotheses does not contain the truth, than these methods should reject all hypotheses. This stands in contrast to previous work that makes the "closed-world assumption" and artificially creates information by comparing measured evidence to synthetic measurements based upon known assumptions. For example, if a multiple-model method is implemented, and the set of hypotheses does not contain the truth, the best one could say is that the hypothesis with the smallest residuals is the most likely hypothesis of all of those tested, based on the available data.

The specific metrics outlined in this work include a measurement dissimilarity metric, which computes the time integral of the error between the estimated target space-object boresight and the line-of-sight vector to each hypothesized target. To discriminate between two closely colocated targets (i.e. those with nearly parallel line-of-sight vectors), a pointing quality metric is presented using the estimates of body angular velocity. This metric computes the Mahalanobis distance using the error between the mean vector of the posterior body angular velocity distribution and each hypothesized body angular velocity tuple. Because some knowledge of the SO shape model is required for the Rao–Blackwellized particle filter used in this work, SO shape model information can be fused with the principles of radiometry to determine maximum body angular velocity errors that satisfy signal-to-noise ratio detection requirements. Finally, the estimated body angular velocities are used to compute mass-specific rotational angular momentum and mass-specific rotational kinetic-energy analogs. These analogs are used in a statistical hypothesis testing framework to systematically eliminate those hypotheses that are not consistent with observational data.

II. Background on Light Curve Inversion

The classification methodologies presented in this work rely on estimated distributions of the attitude and body angular rate states. Although a variety of technologies could yield these answers, the specific results presented in this work use light curve inversion. A light curve is a temporally resolved sequence of power measured over a specified bandwidth. Each point in the light curve is the total amount of power reflected by the SO measured from an unresolved image. Because the total amount of power reflected by the SO is dependent on the SO shape and attitude, estimating either the attitude or shape of the SO is possible using the observed light curve [20]. This process is referred to as light curve inversion, and it was initially developed to characterize asteroids [21].

The first work outlining the theoretical application of light curve inversion to SO characterization was given by Hall et al. in 2005 [20]. Sequential filtering techniques were first applied to simulated, nonmaneuvering SOs in 2007. The approach used unscented Kalman filters (UKFs) to estimate either the SO attitude, shape model, or both simultaneously [5]. More sophisticated methods for estimating the SO shape model have been proposed using multiple-model adaptive estimation techniques. A bank of UKFs, each with a different shape model, was implemented to estimate the position, attitude, and attitude rates of a simulated nonmaneuvering SO [14]. Recognizing that UKFs are inappropriate for substantially non-Gaussian distributions, which can be caused by bidirectional reflectivity distribution function (BRDF) measurement models that are nonlinear functions of attitude states, recent efforts used particle filters (PFs) to estimate the attitude states of agile SOs. It was also shown how shape model bias can be included in the estimated states. To account for the unknown SO torques and inertia properties, SO angular rates were modeled as a white noise process [22]. Realizing that including bias states increased the state dimensionality, more recent work introduced an adaptation of the Singer dynamics model to represent the motion of agile SOs. This enabled the now linear body angular velocity states to be analytically marginalized out to reduce the computational burden of a traditional particle filter. This concept, often referred to as Rao–Blackwellization, enables the estimation of attitude and angular velocity states of maneuvering space objects without a priori knowledge of initial attitude while maintaining computational tractability. The Rao–Blackwellized particle filter (RBPF) simulated results showed, for the first time, the full three-degree-of-freedom estimation of an agile SO [17].

This section briefly summarizes the main techniques and assumptions of that work that are relevant to the methodologies presented here. The first subsection outlines the dynamics model that is used to track the motion of an agile SO. The second section outlines the Rao–Blackwellized particle filter that is implemented to estimate the requisite posterior distributions for the classification methodologies.

A. Exponentially Correlated Angular Velocity Model

For an agile SO, the light curve inversion problem involves estimating both attitude and angular velocity states where attitude kinematics are well known but the angular velocity dynamics may be unknown due to unknown control torques, unknown inertia tensors, and unknown disturbance torques [22]. The true state dynamics for SO rotational motion is given by the Euler angle kinematic relationship and Euler's rotational equation of motion [23]:

$$\begin{bmatrix} \dot{\boldsymbol{\theta}}_{\mathcal{N}}^{\mathcal{B}} \\ \dot{\boldsymbol{\omega}} \end{bmatrix} = \begin{bmatrix} \boldsymbol{B}(\boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}})\boldsymbol{\omega} \\ -\boldsymbol{J}^{-1}(\boldsymbol{\omega} \times \boldsymbol{J}\boldsymbol{\omega}) + \boldsymbol{J}^{-1}T \end{bmatrix} \tag{1}$$

The symbol $\boldsymbol{B}(\boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}})$ defines the nonorthogonal relationship between the body angular rates and the attitude coordinates used to represent SO(3) in the inertial frame [23]:

$$\boldsymbol{B}(\boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}}) = \frac{1}{\cos\theta_2} \begin{bmatrix} 0 & \sin\theta_3 & \cos\theta_3 \\ 0 & \cos\theta_2\cos\theta_3 & -\cos\theta_3\sin\theta_3 \\ \cos\theta_2 & \sin\theta_2\sin\theta_3 & \sin\theta_2\cos\theta_3 \end{bmatrix} \tag{2}$$

The system state is defined as $\boldsymbol{x}^T = [\boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}T}\boldsymbol{\omega}^T]^T$, where $\boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}}$ are the 3-2-1 Euler angles defining the rotation between the SO body frame $\mathcal{B}$ to the inertial frame $\mathcal{N}$, and the body angular velocity of the SO is denoted by $\boldsymbol{\omega}$. Here, $\boldsymbol{J}$ and T represent the true inertia tensor matrix and the sum of all applied control and external torques, respectively. For agile SOs, the true inertia and torques acting on a SO are typically unknown.

To model the unknown inertia matrix and torques acting on an agile SO, an exponentially correlated angular acceleration dynamic model is adopted. This leads to the continuous dynamics model proposed in this work for agile SOs:

$$\begin{bmatrix} \dot{\boldsymbol{\theta}}_{\mathcal{N}}^{\mathcal{B}} \\ \dot{\boldsymbol{\omega}} \\ \dot{\boldsymbol{\alpha}} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{3\times3} & \boldsymbol{B}(\boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}}) & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathcal{I}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} & \mathrm{diag}(-\boldsymbol{\beta}) \end{bmatrix} \begin{bmatrix} \boldsymbol{\theta}_{I}^{B} \\ \boldsymbol{\omega} \\ \alpha \end{bmatrix} + \begin{bmatrix} \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} \\ \mathcal{I}_{3\times3} \end{bmatrix} \boldsymbol{w} \tag{3}$$

The body angular velocity is defined as $\boldsymbol{\omega}$, and $\boldsymbol{\alpha}$ is the body angular acceleration. The process noise, used to simulate the unknown motion of the SO, is denoted as $\boldsymbol{w}$ and driven by the power spectral density of the angular acceleration, as given by Eq. (4):

$$Q_\alpha(\tau) = 2\beta\sigma_m^2\delta(\tau)\mathcal{I}_{3\times3} \tag{4}$$

To implement the model in discrete time, the spectral density matrix must be related to the discrete time, exponentially correlated, process noise. The resultant discrete process noise for the attitude and angular velocity is given [24]:

$$\boldsymbol{Q}_k(t_{k+1},t_k,\boldsymbol{x}) = \int_{t_k}^{t_{k+1}} \boldsymbol{\Phi}(t_{k+1},\mathfrak{s};\beta)\boldsymbol{G}(\mathfrak{s})\boldsymbol{Q}_\alpha(\tau)\boldsymbol{G}(\mathfrak{s})^T\boldsymbol{\Phi}^T(t_{k+1},\mathfrak{s};\beta)\,\mathrm{d}\mathfrak{s} \tag{5}$$

where $\boldsymbol{\Phi}(t_{k+1},\mathfrak{s};\beta)$, which is the state transition matrix, is computed using numerical integration of the continuous time state matrix given in Eq. (3).

The success of this model lies in the successful determination of β and σ_m^2. Like previous models, Q_α, by careful selection of β and σ_m^2, must be sized appropriately to represent the SO maneuver capability. Because β is the inverse of τ, it is selected by matching τ to the length of the expected SO maneuver. It is important to note that, when $\tau \neq 0$, the angular velocities are continuous in time, as they are in the true kinematic motion of the SO. Additionally, as $\tau \to \infty$, the process noise is described as a random walk process. In the special case where $\tau = 0$, the dynamics reduce to a white noise process model very similar to that proposed by Holzinger et al. [22].

Singer proposed modeling the variance of target acceleration as a ternary uniform distribution, where probabilities of success and failure of the maneuver were assigned [24]. This work avoids assuming these unknown probabilities by modeling the variance in the acceleration as a uniform distribution having a maximum acceleration amplitude $\alpha_{\max}$ as defined by Eq. (6):

$$\sigma_m^2 = \frac{\alpha_{\max}^2}{3} \tag{6}$$

The motivating reason for assuming a uniform distribution is the lack of information on SO maneuvering tendencies. Therefore, it is assumed that any magnitude maneuver executed by the SO is equally likely up to the maximum capability of the system defined by $\alpha_{\max}$. Because $\alpha_{\max}$ will seldom be known, one has two primary methodologies for establishing this maximum value. The first is to pick values representative of the hypothesized SO class, where knowledge of SO size could come from auxiliary characterization. The second methodology is to hypothesize multiple targets the SO could be tracking and selecting the maximum acceleration of those hypotheses.

B. Rao–Blackwellized Particle Filter

Although several correlation functions are available to model the dynamics of an agile SO, the exponentially correlated angular acceleration model is chosen purposefully. In Eq. (3), the Euler angles appear nonlinearly in the dynamics and measurement models, whereas the angular velocity and angular acceleration states appear linearly in the dynamics. The linear structure of the angular velocity and acceleration states can be exploited to reduce the computational burden of a traditional particle filter by marginalizing out the linear state variables. These linear states can then be estimated using a Kalman filter (KF), which improves state estimates because it is the optimal estimator of these linear states. This concept is sometimes referred to as Rao–Blackwellization, or as a marginalized filter [25]. Using the notation of Schon et al., the state space is segregated into a nonlinear portion $\boldsymbol{x}_k^n$ and a linear portion $\boldsymbol{x}_k^\ell$, as illustrated in Eq. (7):

$$\boldsymbol{x}_k = \begin{bmatrix} \boldsymbol{\theta}_{\mathcal{N}}^{\mathcal{B}} \\ \hline \boldsymbol{\omega} \\ \boldsymbol{\alpha} \end{bmatrix}_{t=t_k} = \begin{bmatrix} \boldsymbol{x}_k^n \\ \hline \boldsymbol{x}_k^\ell \end{bmatrix} \tag{7}$$

The goal of a nonlinear non-Gaussian filter, like the Rao–Blackwellized particle filter, is to determine the posterior probability density function $p(\boldsymbol{x}_k|\boldsymbol{Y}_k)$ recursively. The RBPF approach analytically marginalizes out the linear state by solving for the posterior probability density function (PDF) at time step k as

$$p(\boldsymbol{x}_k^n,\boldsymbol{x}_k^\ell|\boldsymbol{Y}_k) = \underbrace{p(\boldsymbol{x}_k^\ell|\boldsymbol{x}_k^n,\boldsymbol{Y}_k)}_{\text{KF}}\underbrace{p(\boldsymbol{x}_k^n|\boldsymbol{Y}_k)}_{\text{PF}} \tag{8}$$

where $p(\boldsymbol{x}_k^\ell|\boldsymbol{x}_k^n,\boldsymbol{Y}_k)$ is the marginalized posterior probability density for the linear states and can be optimally solved with the Kalman filter. However, the marginalized posterior probability density for the nonlinear states $p(\boldsymbol{x}_k^n|\boldsymbol{Y}_k)$ has no closed-form solution; therefore, it is approximated with a particle filter [26]. The posterior for the linear portion of the state space is given by

$$p(\boldsymbol{x}_k^\ell|\boldsymbol{x}_k^n,\boldsymbol{Y}_k) = \mathcal{N}(\hat{\boldsymbol{x}}_k^\ell,\boldsymbol{P}_k) \tag{9}$$

where $\hat{\boldsymbol{x}}_k^n$ and $\boldsymbol{P}_k$ are the linear mean and covariance determined from a KF with the nonlinear states marginalized. Combining the KF posterior with the PF posterior gives the overall state posterior PDF as

$$p(\boldsymbol{x}_k^n,\boldsymbol{x}_k^\ell|\boldsymbol{Y}_k) = \sum_{i=1}^{N} p(\boldsymbol{x}_k^\ell|\boldsymbol{x}_k^n(i),\boldsymbol{Y}_k)w_k^i\delta(\boldsymbol{x}_k^n - \boldsymbol{x}_k^n(i)) \tag{10}$$

Note that this representation of the PDF can model both non-Gaussian PDFs in the linear and nonlinear states. In this representation, the linear states are modeled using a Gaussian distribution, assuming the initial state is Gaussian, and the nonlinear states are modeled using a collection of weighted particles. Previous work explored in the greater detail the implementation of the RBPF to this problem [17]. To briefly summarize, the measurements of the target SO y_k are defined by the photon flux captured by the optical system q_{SO}, measured in electrons/second:

$$y_k(\boldsymbol{x},t_k) = q_{\mathrm{SO}}(\boldsymbol{x},t_k) + v_k(\boldsymbol{x},t_k) \tag{11}$$

$$v_k(\boldsymbol{x}, t_k) \sim \mathcal{N}(\boldsymbol{0}, R_k(\boldsymbol{x}, t_k)) \tag{12}$$

The function used to evaluate the likelihood of each particle $\tilde{c}_k^i$, conditioned on the true measurement y_k, is defined by Eq. (13):

$$\begin{aligned}\tilde{c}_k^i &= p(z_{1,k}|\boldsymbol{x}_k) \\ &= \frac{1}{(2\pi R_k(\boldsymbol{x}_k, t_k, t_I))^2} \exp\left[-\frac{1}{2}(z_{1,k}^i R_k(\boldsymbol{x}_k, t_k, t_I)^{-1} z_{1,k}^i)\right]\end{aligned} \tag{13}$$

Here, the quantify $z_{1,k}^i$ is the measurement residual:

$$z_{1,k}^i = y_k - h_k^n(\boldsymbol{x}_k^{i,n}, t) \tag{14}$$

These likelihoods are often referred to as the "importance weights" of each particle, and they are used in the PF resampling algorithm after normalizing the weights. This approach uses systematic resampling [26], although other methods such as Metropolis or stratified resampling have been offered as equally effective alternatives [27].

III. Methodology

The previous section outlined the light curve inversion approach developed for agile SOs. This section presents several inference techniques for determining the most likely subject under observation by an agile SO. To give context to these contributions, it is prudent to discuss the spectrum of SO characterization. Consider Fig. 1, which proposes three general levels of SO classification. The general concept conveyed by Fig. 1 is that, as information is gathered on a SO, further characterization is enabled.

The most basic characteristics of SOs, such as size or shape, could be estimated with no a priori information available [4]. Typically, the characteristics determined are intrinsic properties of the SO; therefore, this level constitutes a physical characterization. Determining the shape model enables light curve inversion techniques, affording estimates of current SO attitude or enabling inference on SO active control mode. Therefore, this information constitutes a functional description of current SO activity. Although methods for determining the shape-independent attitude of nonagile SOs have been presented [3], the current work focuses on agile SOs. Finally, if these activities are aggregated over time, further inference could help establish routine operational patterns or indicate the payload capability of the SO.

The contributions of this work best fit under the "functional characterization" block of Fig. 1. Sections III.A and III.B outline metrics that use the estimated states directly to infer the active control mode of the SO. Section III.C couples statistical inference techniques with angular momentum and rotational kinetic-energy analogs to establish formal hypothesis testing techniques. Finally, Sec. III.D outlines how radiometric first principles can bound the allowable tracking error between the target and the subject SOs.

A. Inference Using Attitude States

As illustrated in Fig. 2, $\boldsymbol{\rho}_{ij}$ defines the line-of-sight vector from the SO to a hypothesized target, and $\hat{\rho}_b$ defines the vector along the SO boresight calculated from the estimated attitude angles. The simplest technique to infer the SO active control mode is to determine the smallest angle between the estimated attitude states and each hypothesized attitude tuple. The instantaneous angle between two unit vectors, $\hat{\boldsymbol{\rho}}_b$ and $\hat{\boldsymbol{\rho}}_{ij}$, is determined from the definition of the cross product as shown in Eq. (15):

$$\Delta\theta_{a,b}(t) = 2 \ \arcsin\left(\frac{\|\hat{\boldsymbol{\rho}}_b(t) \times \hat{\boldsymbol{\rho}}_{ij}(t)\|}{2}\right) \tag{15}$$

Intuitively, the most likely subject of observation for the target SO at any instant in time is the hypothesis with the smallest instantaneous angle. However, for very cluttered regions of space, such as those regions near to the poles, it is the case that many hypothesized SOs have small instantaneous angles.

To address this problem, this work adopts a measurement arc dissimilarity metric (MDM) developed for a match filtering technique for detecting SOs in low signal-to-noise ratio imagery [16]. To classify the active control mode of the SO, one desires the dissimilarity in the measurement arcs over the entire observation window $t \in [t_k - T, t_k]$. For ease of interpretation, one can define the mean MDM as given by Eq. (16):

$$\bar{d}_{\mathrm{MD}} = \frac{1}{T}\int_{t_k-T}^{t_k} \Delta\theta_{a,b}(\tau)\,\mathrm{d}\tau \tag{16}$$

Because the bidirectional reflectivity distribution function measurement models of SOs are nonlinear functions of attitude states, non-Gaussian state distributions frequently arise. Commonly, the particle cloud grows to define a multimodal distribution representing the symmetry commonly exhibited by manmade satellites. For example, if a satellite shaped as a cube has roughly the same material properties on each side, the RBPF correctly exhibits distinct clusters of particles. As shown in Fig. 3, it is common for the mean attitude state, represented by the open circle, to lie outside the actual particle cloud. To more closely align the vectors $\hat{\boldsymbol{\rho}}_{ij}$ and $\hat{\boldsymbol{\rho}}_b$, k-means++ clustering is used to identify the number of distinct clusters in the particle cloud. However, it should be noted that this clustering step is not necessary.

In traditional k means, as developed by Lloyd, an arbitrary number of data samples (in this case, particles of a PF) are assigned to c clusters where the number c is selected by the analyst [28]. Although the methodologies presented in this work are not developed specifically with automation in mind, the more sophisticated k-means ++ algorithm, such as implemented in MATLAB®, does not require a priori specification of c [29]. Additionally, there is no theoretical reason why other clustering or fuzzy logic routines would be less appropriate for the task at hand. Moreover, it may serve as motivation for the implementation of a Rao–Blackwellized unscented Kalman filter, as discussed in the previous section.

Figure 3 illustrates an instantiation of the attitude states at a single time step that resulted in two distinct clouds, shaded lighter and darker gray, respectively. Because each particle has an associated weight, it is possible to identify the most likely cluster of particles by summing all the weights in each identified cluster. The weights themselves are not used in the k-means++ algorithm to form each cluster. The centroid of that cluster is then taken as $\hat{\rho}_b$, shown as the black X in Fig. 3.

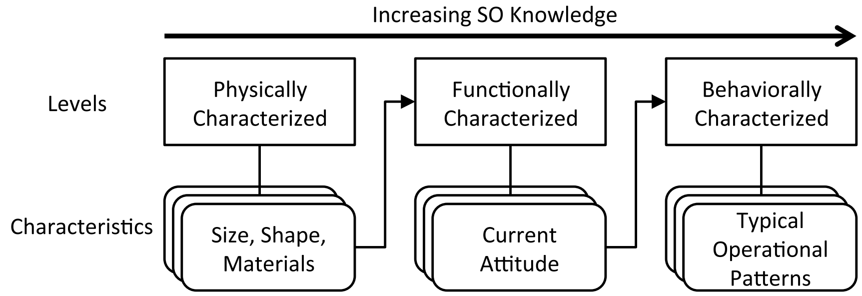
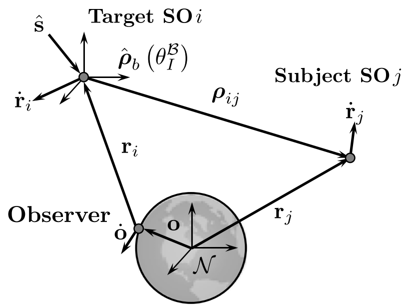
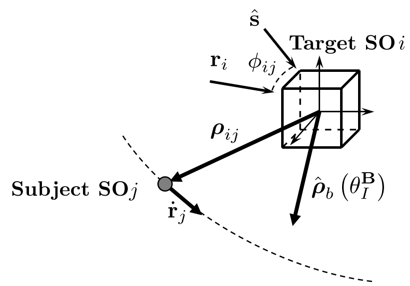


Fig. 1 Increasing knowledge of agile SO enables refined characterization.



a) Observer, target SO, and subject SO geometry



b) Instantaneous angle between target sensor bore axis and subject SO

Fig. 2 Problem geometry and instantaneous angle illustration.

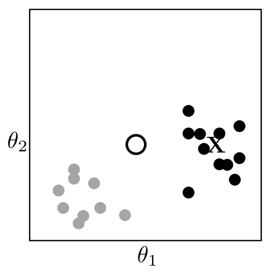


Fig. 3 2-D view of attitude states sorted via k means.

If the target SO continuously tracks the same subject, the correct hypothesis is described by the lowest-magnitude MDM. This will not be the case if the target SO exhibits discrete changes in attitude during the observation window. It is also possible that hypotheses with different orbits appear to be closely oriented in space from the perspective of the target SO. As an intuitive approach to disambiguate these important cases, one could examine the time derivative of the instantaneous angle defined by Eq. (15) to provide more information:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\Delta\theta_{a,b}(t)) = \frac{\delta}{\delta t}\left(2\arcsin\left(\frac{\|\hat{\boldsymbol{\rho}}_b(t)\times\hat{\boldsymbol{\rho}}_{ij}(t)\|}{2}\right)\right) \tag{17}$$

The final derivative to be evaluated has previously been derived, as shown in Eq. (18) [30]:

$$\left\|\frac{^{\mathcal{B}}\mathrm{d}\hat{\boldsymbol{\rho}}_{ij}(t)}{\mathrm{d}t}\right\| \equiv \omega_\ell = \left\|\frac{1}{\|\hat{\boldsymbol{\rho}}_b\|}\left[\mathcal{I} - \frac{\hat{\boldsymbol{\rho}}_b\hat{\boldsymbol{\rho}}_b^T}{\hat{\boldsymbol{\rho}}_b^T\hat{\boldsymbol{\rho}}_b}\right](\dot{\boldsymbol{r}}_j - \dot{\boldsymbol{r}}_i) + \boldsymbol{\omega}_{\mathcal{N}}^{\mathcal{B}} \times \left[\frac{\boldsymbol{r}_j - \boldsymbol{r}_i}{\|\boldsymbol{r}_j - \boldsymbol{r}_i\|}\right]\right\| \tag{18}$$

Examining Eq. (18) reveals that there are two main contributing factors to the change in angle between the target SO sensor boresight and the subject SO. The first term on the right-hand side of Eq. (18) is simply dependent on the inherent motion of the target and the subject SOs. The second term describes the component of angle change that is dependent on the rotational motion of the agile SO $\boldsymbol{\omega}_{\mathcal{N}}^{\mathcal{B}}$. Fortuitously, the RBPF previously developed by the authors and their colleagues enables estimation of the body angular velocities of the target SO $\boldsymbol{\omega}_{\mathcal{N}}^{\mathcal{B}}$ to discriminate between subjects that are colocated along the target SO boresight $\hat{\boldsymbol{\rho}}_b(t)$. Therefore, the remaining techniques in this work seek to quantify the pointing quality using the estimated posterior body angular velocity distributions of the target SO. If it is found that the body angular velocity of the target SO matches a profile necessary to track a specific subject SO, the observer may infer intentional action by the target SO.

B. Ranking Hypotheses Using Mahalanobis Distance

The most general expression for the inner-product distance between two vectors, $\boldsymbol{a}$ and $\boldsymbol{b}$, is given by

$$d(\boldsymbol{a},\boldsymbol{b})^2 = (\boldsymbol{a}-\boldsymbol{b})^T\boldsymbol{A}^{-1}(\boldsymbol{a}-\boldsymbol{b}) \tag{19}$$

Here, the distance matrix $\boldsymbol{A}$ is positive definite. When $\boldsymbol{A} = \mathcal{I}$, the Euclidean distance formula is recovered, whereas setting it equal to the covariance matrix of a Gaussian distributions $\boldsymbol{A} = \boldsymbol{P}$ yields the Mahalanobis distance.

The particle cloud representing the body angular velocity states can be expressed as a multivariate Gaussian mixture model (MGMM). Generally speaking, an MGMM expresses a probability density function as a weighted sum of $i = 1, \ldots, N$ Gaussian component densities, as shown in Eq. (20):

$$p(\boldsymbol{x}_k^\ell) = \sum_{i=1}^{N} w_i f_i(\boldsymbol{x}) \tag{20}$$

Here, w_i is referred to as a "weight" or "likelihood" in the estimation literature. The RBPF represents the distribution with N "particles", as they are referred to in the estimation literature, or as "component densities" in the statistics literature. Each component density is a p-variate Gaussian function of the form

$$f(\boldsymbol{x}) = \mathcal{N}(\boldsymbol{x}_k^\ell|\boldsymbol{\omega}_{\mu,i},\boldsymbol{P}_i) \tag{21}$$

To evaluate the distance of multivariate Gaussian mixture models, alternative expressions for the covariance matrix are required. Previous work has advocated for a Gaussian mixture distance function based on minimizing the Kullback–Leibler (KL) divergence [31]. Minimizing the KL divergence for a Gaussian mixture yields the distance matrix given by Eq. (22):

$$\boldsymbol{A}(\boldsymbol{x}) = \left[\left(\sum_{i=1}^{n} w_i f_i(\boldsymbol{x})\boldsymbol{P}_i\right)^{-1}\left(\sum_{j=1}^{n} w_j f_j(\boldsymbol{x})\boldsymbol{P}_j\right)\right]^{-1} \tag{22}$$

Here, $\boldsymbol{A}(\boldsymbol{x})$ is the weighted reciprocal sum of the component covariances, and the weights are those given by the RBPF. To express the Mahalanobis distance for an MGMM, one also must define the mean of the MGMM. The mean of a MGMM is given simply by the weighted sum of all components:

$$\boldsymbol{\omega}_\mu = \sum_{i=1}^{n} w_i\boldsymbol{\omega}_{\mu,i} \tag{23}$$

Substituting both Eqs. (22) and (23) into Eq. (19) yields the Mahalanobis distance analog for a MGMM. In this work, the Mahalanobis distance analog is evaluated for each hypothesized subject ($h = [1, \ldots, M]$ at each time interval), such that

$$d_{\mathrm{GM}} = \sqrt{(\boldsymbol{\omega}_{h,k} - \boldsymbol{\omega}_{\mu,k})^T\boldsymbol{A}^{-1}(\boldsymbol{\omega}_{h,k} - \boldsymbol{\omega}_{\mu,k})} \tag{24}$$

where $\boldsymbol{\omega}_{h,k}$ is the body angular velocity tuple for each hypothesis $H_{h,k}$ at each time step t_k. The Mahalanobis distance can then be computed

Algorithm 1 Rank hypotheses using Mahalanobis distance

<u>function rank MD</u> ($\boldsymbol{\omega}_{\mu,k}, \boldsymbol{\omega}_{h,k}, \boldsymbol{A}(\boldsymbol{x})$);
Input: $\boldsymbol{\omega}_{\mu,k}, \boldsymbol{\omega}_{h,k}, \boldsymbol{A}(\boldsymbol{x})$
Output: flag
For each time step;
for $k \leftarrow 1$ **to** T, **do**
  For each hypothesis i;
  for $h \leftarrow 1$ **to** M, **do**
    Evaluate Mahalanobis distance, Eq. (24);
    $d_{\mathrm{GM}}(h,k) = \sqrt{(\boldsymbol{\omega}_{h,k} - \boldsymbol{\omega}_{\mu,k})^T \boldsymbol{A}^{-1} (\boldsymbol{\omega}_{h,k} - \boldsymbol{\omega}_{\mu,k})}$;
  end
end

for each hypothesis at each time step, where the hypothesis with the lowest value of Mahalanobis distance (MD) indicates the most likely hypothesis, as shown in Algorithm 1.

C. Angular Momentum and Rotational Kinetic-Energy Analog Inference

The first principle tells us that the momentum and energy states of a SO are conserved quantities, unless acted upon by an outside force [32]. Therefore, changes in either angular momentum or angular kinetic energy could serve as a basic indication of the active control mode. In this work, the inertia tensor of the space objects is assumed to be constant but unknown. Therefore, this work proposes analogs for the angular momentum and angular kinetic energy as given by Eqs. (26) and (28):

$$\boldsymbol{H} = \boldsymbol{J}\boldsymbol{\omega} \tag{25}$$

$$J_H = \boldsymbol{\omega} \tag{26}$$

$$E = \frac{1}{2}\boldsymbol{\omega}^T \boldsymbol{J}\boldsymbol{\omega} \tag{27}$$

$$J_E = \boldsymbol{\omega}^T \boldsymbol{\omega} \tag{28}$$

As before, $\boldsymbol{J}$ is the unknown inertia matrix and $\boldsymbol{\omega}$ is the body angular velocity, which are both of the target SOs. Here, J_H is the angular momentum analog, which is a rotated and skewed version of the true angular momentum. J_E is the rotational kinetic-energy analog, which is simply proportional to and monotonic with the true rotational kinetic energy. These definitions imply that J_H is defined as an MGMM and J_E is defined as an MGMM analog of the chi-squared distribution with 3 deg of freedom:

$$J_H = w_i \sum_{i=1}^{N} \mathcal{N}(\boldsymbol{x}_k^{\ell} | \boldsymbol{\omega}_{\mu,i}, \boldsymbol{P}_i) \tag{29}$$

$$J_E = \left[\sum_{i=1}^{N} w_i \mathcal{N}(\boldsymbol{x}_k^{\ell} | \boldsymbol{\omega}_{\mu,i}, \boldsymbol{P}_i)\right]^T \left[\sum_{i=1}^{N} w_i \mathcal{N}(\boldsymbol{x}_k^{\ell} | \boldsymbol{\omega}_{\mu,i}, \boldsymbol{P}_i)\right] \tag{30}$$

From a geometric standpoint, the angular velocity vector is constrained to lie on the intersection of the angular momentum ellipsoid and the kinetic-energy ellipsoid. The path defined by this intersection is called the "polhode," and it defines the curve of the angular velocity vector as seen from the body-fixed frame. Assuming ideal rigid bodies with no internal energy dissipation, the true angular momentum and rotational kinetic energy are conserved, and the polhode forms a closed path [33].

More important, the magnitude of the angular momentum analog is constant under torque-free motion for axisymmetric rigid bodies. By using the outputs of the RBPF to build momentum and energy distributions, statistical tests can be used to determine the active control mode of an agile SO. Generally, these tests could provide evidence of the following behaviors: whether the SO is actively maneuvering during the observed period, what subject a target SO is observing, and whether the active control mode of the target SO changes between any two arbitrary points in time. Consequently, the authors propose using multivariate hypothesis tests to match the estimated SO momentum or energy analog with momentum or energy profiles for each hypothesized subject SO.

To briefly review, a hypothesis is a claim about a certain statistical parameter, where the union of the null hypothesis H_0 and alternative hypothesis H_1 comprise the entire domain of the parameter. Of these two, the null hypothesis is the "status quo" and the alternative hypothesis is the new hypothesis being tested. To demonstrate the use of the momentum analog, a two-sided hypothesis test is implemented [34]. Dropping the time-step k notation, the form of the hypothesis is given by Eq. (31):

$$H_0{:}J_H = J_{H,h} \text{ vs. } H_1{:}J_H \neq J_{H,h} \tag{31}$$

This form is chosen because it is irrelevant whether the angular momentum profile of the target SO is larger or smaller than any chosen hypothesized momentum profile. The goal is simply to determine which hypotheses are statistically equivalent to the estimated values. The statistical hypothesis test for the equality of two means requires that the two distributions are Gaussian. To test the mean vector of a multivariate Gaussian distribution, the relevant test statistic is T^2, which is also referred to as Hotelling's T^2 in the social sciences [35].

$$T^2 = N(\boldsymbol{\omega}_{\mu,k} - \boldsymbol{\omega}_{h,k})^T \boldsymbol{A}(\boldsymbol{x})^{-1} (\boldsymbol{\omega}_{\mu,k} - \boldsymbol{\omega}_{h,k}) \tag{32}$$

Here, N is traditionally the number of samples drawn from the true population. In this work, N is then the number of particles used in the RBPF. The null hypothesis is rejected when the test statistic is greater than a critical value ($T^2 > T^2_{\mathrm{crit}}$), where the critical value is

$$T^2_{\mathrm{crit}} = \frac{(N-1)p}{N-p} F_{p,N-p}(\alpha) \tag{33}$$

Here, F is the F-cumulative distribution function with p and $N-p$ degrees of freedom, respectively. Recall that each N is the number of particles, or component densities, and that each component density is a Gaussian function length of dimension p. This cumulative distribution function is evaluated at the user-selected value of α, which is the probability of a type 1 error, and is referred to as the "level of significance" of the test. Type 1 and type 2 errors are two examples of incorrect conclusions that could be reached by a hypothesis test. A type 1 error indicates that H_0 is actually true but that H_0 is incorrectly rejected. A type 2 error indicates that H_0 should have been rejected but the hypothesis test failed to reject H_0.

Although the T^2_{crit} critical region works well for the J_H analog as it approaches normality, sufficiently non-Gaussian distributions, such as J_E, are ill-suited for these tests. Because the number of particles N is typically large in the RBPF, one can also construct the hypotheses tests with confidence regions that do assume normal distributions. These large sample inferences are made using the chi-squared distribution, with the critical value shown in Eq. (31) [35]:

$$\chi^2_{\mathrm{crit}} = \chi^2_p(\alpha) \tag{34}$$

Here, $\chi^2_p(\alpha)$ is the αth upper percentile of the chi-squared distribution with p degrees of freedom. This leads to Algorithm 2, which systematically eliminates those hypotheses that are not consistent with observational data at each time step k.

D. Bounding Tracking Error with Detection Statistics

The appeal of the metrics presented in the previous section is that no additional information is required beyond that which is provided by the RBPF algorithm. However, shape model information is necessary for the light curve inversion process, meaning that the

Algorithm 2 Momentum and energy inference

<u>function momentum inference</u> $(\boldsymbol{\omega}_{\mu,k}, \boldsymbol{\omega}_{h,k}, \boldsymbol{A}(\boldsymbol{x}), \alpha)$;
Input: $\boldsymbol{\omega}_{\mu,k}, \boldsymbol{\omega}_{h,k}, \boldsymbol{A}(\boldsymbol{x}), \alpha$
Output: flag
For each time step;
for $k \leftarrow 1$ **to** T, **do**
  For each hypothesis i;
  for $h \leftarrow 1$ **to** M, **do**
    Evaluate test statistic, Eq. (32);
    $ts \leftarrow N(\boldsymbol{\omega}_{\mu,k} - \boldsymbol{\omega}_{h,k})^T \boldsymbol{A}(\boldsymbol{x})^{-1} (\boldsymbol{\omega}_{\mu,k} - \boldsymbol{\omega}_{h,k})$
    Determine critical region, Eq. (34);
    $\chi^2_{\mathrm{crit}} \leftarrow \chi^2_p(\alpha)$
    if $ts > \chi^2_{\mathrm{crit}}$, **then**
      Reject H_0 at α level of significance;
      flag$\leftarrow$0
    else
      Fail to reject H_0 at α level of significance;
      flag$\leftarrow$1
    end
  end
end

general size of the target SO is known. Therefore, it is possible to bound the maximum aperture diameter of the target SO optical system. If it is further assumed that the target satellite is attempting to track its subjects with an electrooptical sensor of its own; this information can be coupled with the statistics of the signal-to-noise ratio to determine a maximum relative angular velocity error that would yield detections of the hypothesized subject SOs. The simulated results also assume that this sensor is not individually articulated, and that the whole target SO maneuvers to track its subject. Finally, this specific formulation presented in this work further assumes that the target is taking unresolved images of the hypothesized subjects, but this assumption could be relaxed by replacing the signal-to-noise ratio equation used.

For ease of illustration, the GMM of a single body axis is represented in Fig. 4. In the figure, $\phi_{\mu,k}$ is the estimated mean of the body angular velocity along the X axis at the kth time step and $\phi_{h,k}$ is the body angular velocity required at that time step to observe the hth hypothesized subject. Please note that the univariate case is illustrated for ease of explanation only, as any mathematical technique should use the full MGMM.

By subtracting each hypothesized body angular velocity from the distribution, one can define a distinct distribution for each hypothesized subjection satellite. Each body angular velocity error distribution is given at each time step by

$$\boldsymbol{e}_{h,k} = \boldsymbol{\omega}_{\mu,k} - \boldsymbol{\omega}_{h,k} \tag{35}$$

Each error PDF can then in turn be used to construct a cumulative distribution function of body angular velocity errors, as shown in Fig. 5. To determine the probability that each hypothesized subject is under observation by the target, one determines the percentage of the error distribution for which the particles had angular velocities less than the limit ω_ℓ. Please note that ω_ℓ is exactly equal to $\mathrm{d}\hat{\boldsymbol{\rho}}_{ij}(t)/\mathrm{d}t$ [introduced in Eq. (18)] and is substituted for brevity of notation.

It is known that, as the angular rate error grows larger, the unresolved "point" of light grows into a streak. To quantify the angular error limit ω_ℓ that demarcates usable data, one needs to relate this angular error and the quality of the unresolved image. One of the most commonly used metrics for expressing the quality of an image capturing is the signal-to-noise ratio. Mathematically, it can be expressed as shown in Eq. (36), which is commonly referred to as the "charge-coupled device (CCD) equation":

$$\mathrm{SNR} = \frac{q_{\mathrm{SO}} t}{\sqrt{q_{\mathrm{SO}} t + m(1 + (m/z))\left[(q_{p,\mathrm{back}} + q_{p,\mathrm{dark}})t + (\sigma_r^2/n^2)\right]}} \tag{36}$$

In Eq. (36), q_{SO} is the photon flux of the subject captured by the target's optical system, t is the integration time of the sensor, m is the number of pixels occupied by the SO, z is the number of pixels used in estimating the background brightness, $q_{p,\mathrm{back}}$ is the average brightness of the background, $q_{p,\mathrm{dark}}$ is the dark current of the sensor, σ_r is the read noise, and n is the binning factor of the CCD. Texts have been devoted to accurately modeling these parameters, but simple estimates are provided by the radiometric model provided by the authors in the appendix of Coder and Holzinger [36]. To implement the SNR, the analyst must appropriately model the brightness of both the subject SO and the background of the image. To see how tracking errors influence the overall SNR of the image, one must examine the quantity m.

When the target is perfectly tracking the subject SO, then the value of m is constant. However, when tracking errors cause a relative velocity between the target's optical system and subject, m will increase monotonically with t. For unresolved images of the subject, the captured image will contain SO streaks. For a SO streaking across the image plane, the number of pixels occupied by the incoming signal grows is defined by Eq. (37):

$$m = m_0 + \frac{\sqrt{m_0}\,\omega_\ell t}{\mathrm{IFOV}} \tag{37}$$

Here, ω_ℓ is the tracking error limit, IFOV defines the angular field of view of a single pixel, and m_i is the number of pixels occupied by the point spread function (PSF) of the point source [37]. The PSF full-width half-maximum (FWHM) of a traditional, terrestrial observatory is influenced by the focus of the optical system, any atmospheric blurring, or the diffraction of light. Because the system under consideration is space based, it is appropriate to use the diffraction limit and assume the optics are well focused.

The PSF due to diffraction limiting is the theoretical limit of an optical system's resolution due to the diffraction of light. For a perfectly circular aperture with no obscuration, the diameter of a point source is described by the "Airy disk." The angle in radians of the Airy disk is given by Eq. (38) [38]:

$$\theta_A = \frac{2.44(\lambda)}{D} \tag{38}$$

It is important to note here that the diameter of the Airy disk is wavelength dependent; therefore, a weighted average for the wavelength should be used.

Although a large number of assumptions needs to be made, the most critical for space-based applications (which is the aperture diameter of the target SO optical payload) can be inferred from

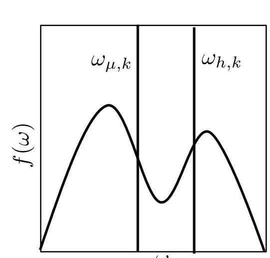


Fig. 4 Direct state-space classification.

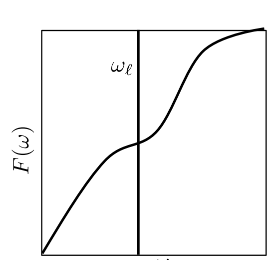


Fig. 5 Cumulative distribution function of error.

Algorithm 3 Bounding tracking error with detection statistics

function bound track error $(\boldsymbol{X}, \boldsymbol{p}, \omega_{\mu,k}, \boldsymbol{\omega}_{h,k})$;
Input:
Output: flag
For each time step;
for $k \leftarrow 1$ **to** T, **do**
  For each hypothesis i;
  for $h \leftarrow 1$ **to** M, **do**
    Bound ω, Eqs. (36–37);
    $\omega_\ell \leftarrow$function SNR$(\boldsymbol{X}, \boldsymbol{p})$
    Determine tracking error distribution, Eq. (35);
    $\boldsymbol{e}_{h,k} \leftarrow \|\boldsymbol{\omega}_{\mu,k} - \boldsymbol{\omega}_{h,k}\|$
    Calculate probability of hypothesis with cumulative distribution function;
    $Pr(\boldsymbol{e}_{h,k} < \omega_\ell) \leftarrow F_{\boldsymbol{e}}(\omega_\ell)$
  end
end

observational data. The aperture diameter is the greatest contributor to limiting magnitude for typical space-based assets [36]. Size estimation techniques routinely provide estimates of the optical cross section and its derivatives in various photometric bands, and it is therefore possible to bound D [3,14,39]. Likewise, the lower bound on focal ratio, or f number, of an optical system is limited by optical aberrations [40].

Although the focus of this work is determining the active control mode of the target SO, these techniques could easily be extended to encompass historically defined operational modes. The term operational mode typically refers to broad categorical behaviors such as "nominal," "anomalous," or "nadir pointing." In the case of non-SO subjects, such as the sun or nadir, these subjects can simply be included as additional hypotheses, as will be demonstrated in the Results section (Sec. IV). Determining anomalous behavior is more involved, as one must baseline which observation subjects are nominal. However, with such a baseline, the determination of subjects different from those composing the baseline is a relatively simple endeavor.

IV. Results

Simulated results for the methodologies outlined in this work are presented in the following. Section IV.A outlines some important considerations for accurately replicating the results presented in this paper. Section IV.B outlines the single test case scenario, and it highlights the results available directly from the RBPF. Section IV.C presents the results of the active control mode analysis using the contributions of this work.

A. Practical Implementation Notes

An important consideration of the methodologies outlined previously is to account for the fact that motion about one or more body axes may be unobservable. This is the case with the example simulation used to produce these results. In the simulated scenario, any arbitrary rotation about the body y axis does not cause any significant change to the visible areas of the target SO. As a result, information on this axis, such as the mean angular velocity, should be excluded. Mathematically, this implies that $p = 2$ degrees of freedom, and that any vector $\boldsymbol{\omega}$ should contain only the roll and yaw axes.

Additionally, simple checks can reduce the initial hypothesis space. The first is to ensure that the target SO can maintain the line of sight with the subject SO. Simple algebraic algorithms have been developed that remain computationally tractable even for a large set of hypothesized subjects. Similar checks exist to ensure that the subject is illuminated with respect to the target SO. The specific line-of-sight algorithm implemented to generate the results here is algorithm 35 from the work of Vallado [41].

Finally, Table 1 summarizes what is both known and unknown about the target SO. Using only measurements of the brightness coupled with knowledge on the shape and material properties of the SO, one is able to estimate the attitude and attitude rate states of a maneuvering satellite. One does not require an initial guess for the orientation of the satellite, the inertia tensor, or the torques applied by actuators on the SO.

Table 1 Known quantities for light curve inversion using RBPF

Quantity	Known	Unknown
Inertia tensor	— —	✓
SO torques	— —	✓
Initial attitude	— —	✓
SO shape model	✓	— —

Table 2 Assumed shape model parameters

Facet	A, m	ξ	a	m
$+X$	2	0.5	0.25	0.3
$+Y$	2	0.5	0.5	0.3
$+Z$	2	0.5	0.75	0.3
$-X$	2	0.5	0.9	0.3
$-Y$	2	0.5	0.4	0.3
$-Z$	2	0.5	0.1	0.3

INTELSAT 18
1 37834U 11056A 15276.58183779 .00000024 00000-0 00000+0 0 9992
2 37834 0.0167 127.6907 0001915 64.8570 208.8878 1.00269536 14658

Fig. 6 Intelsat 18 two-line element set.

B. Simulated Scenario

The simulated results presented assume that the observer is located at the Remote Maui Experiment in Kihei, Hawaii. The telescope parameters used are those of the 0.5 m, $f/8$, Georgia Institute of Technology's Space Object Research *Telescope* [36], and they are representative of a typical Raven-class telescope [42]. The simulated measurements are 1 s exposures collected once every 10 s. The simulated target SO is the Intelsat 18 (IS-18) satellite, where the two-line element set (TLE) shown in Fig. 6 was downloaded from space-track.org.[§] The simulation represents observations collected on 5 October 2015 from 08:05:00 to 08:15:20 Coordinated Universal Time (UTC).

The TLE is ingested into the MATLAB implementation of the simplified general perturbations propagator (referred to as SGP4) developed by Vallado et al. [43]. The ephemerides of the sun and Earth are calculated with less than a 1 in. error using the 1987 implementation of Variations Séculaires des Orbites Planétaires (referred to as VSOP87) [44]. These programs use the mean equator and equinox of J2000 to define the Earth-centered inertial reference frame

The shape model of IS-18 is simplified to that of a simple cube, with the shape model parameters presented in Table 2. This simplification is made to expedite the simulation runs and is not a limitation of the RBPF. The measurements are simulated with a BRDF that uses an affine combination of the Cook–Torrance BRDF model for specular reflectance [45] and Lambertian reflectance for diffuse reflectance. In the shape model, A is the facet area; $\boldsymbol{\xi}$ is the affine transformation weighting parameter; a is the diffuse albedo; and m is the microfacet slope parameter where ξ, a, and $m \in [0, 1]$ [22].

The RBPF is instantiated with 10,000 particles. Because previous work has addressed shape model uncertainty, this work assumes the shape model is known perfectly [22]. To simulate the motion of IS-18, the problem is modeled such that the optics payload is fixed to

[§]Information from Space-Track.org was retrieved 25 November 2015.

Table 3 Hypothesized subject SO

Catalog identification number	Common name	$\overline{MDM}$, deg
16111	SL-3 R/B	8.89
21423	SL-14 R/B	— —
21397	OKEAN-3	9.58
16182	SL-16 R/B	9.70
18187	COSMOS 1867	10.18
19120	SL-16 R/B	8.84
20511	SL-14 R/B	— —
22803	SL-16 R/B	— —
25400	SL-16 R/B	9.49
27422	IDEFIX & ARIANE 42P R/B	11.34
28059	CZ-4B R/B	— —
31793	SL-16 R/B	9.03

the $-Y$ facet of IS-18, which tracks SO 16111 [an SL-3 rocket body (R/B)], for the duration of the pass. Table 3 lists the 12 hypothesized subjects for this simulation, which are all positioned in their orbits such that they are crossing the north pole. It is again emphasized that this set of hypotheses represents a mutually exclusive set, but it is by no means an exhaustive list of potential subjects.

Figure 7 illustrates the problem geometry, where the simulated subject SOs are highlighted, and the true target is denoted. Emphasizing the difficulty of this problem, Fig. 8 illustrates the subjects from IS-18's point of view. It is easily seen how attitude information alone makes inference of the most probable subject difficult.

The attitude states estimated by the RBPF are presented in Figs. 9 and 10. Figure 9 shows the two-dimensional (2-D) view of the roll and pitch axes, and each subfigure is a different time step in the simulation. Figure 10 shows the 2-D view of the roll and yaw Euler angles, where each subfigure again denotes a separate time step. In the figures, the dots are the particles of the RBPF, with each shaded to

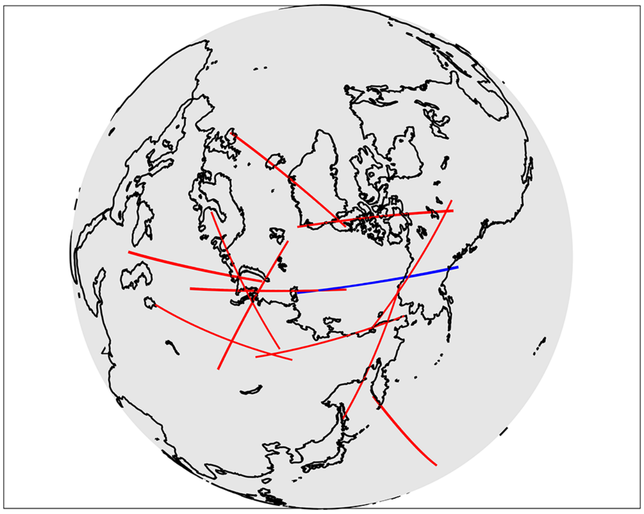


Fig. 7 Top-down view of simulated scenario.

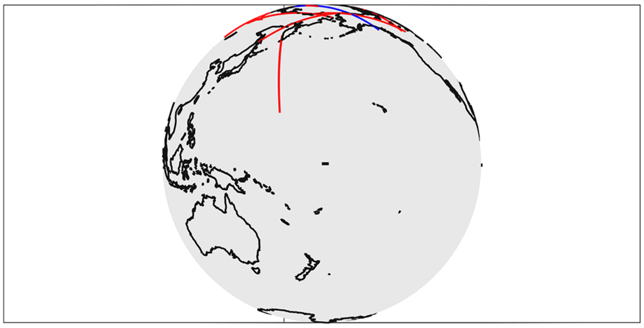


Fig. 8 Depiction of subject SO from perspective of IS-18.

indicate the likelihood measured for that particle. Black indicates the most likely particles, whereas light gray indicates less likely particles. The star indicates the true attitude state of IS-18 necessary to track SO 16111.

Figure 11 represents the same particle cloud and time step as Figs. 9a and 10a, but the Euler angles have been transformed to topocentric equatorial coordinates. The x axis denotes right ascension, the y axis denotes declination, and the origin is fixed to the camera boresight of the target SO. Consequently, the star representing the true target remains fixed at the origin, just above the north pole. The circle is the Earth exclusion zone, the triangle is the centroid of the particle cloud, and the dots represent the incorrect hypotheses. It can be seen that the initial particle cloud, defined as uniform over the Euler angles, initially covers the entire Earth.

Figure 12 represents the same particle cloud and time step as Figs. 9d and 10d, with the same symbology as Fig. 11. It can be seen that the particle cloud is reduced from its initial coverage to a band that spans the poles of the Earth. The geometry of the cloud is likely due to the lack of observability along the y axis as well as the simplistic shape model used for IS-18.

C. Active Control Mode Inference

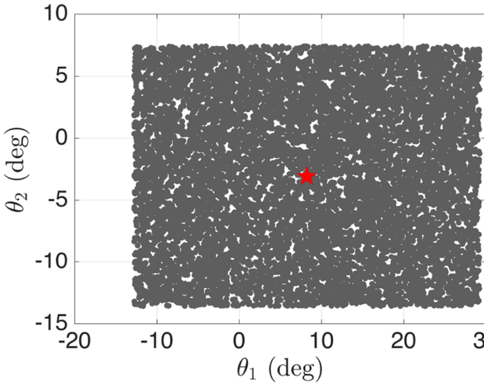
Using these state estimates, the results for the proposed active control mode inference techniques are presented. For the instantaneous angle, the first step in obtaining a more accurate centroid involves the k-means++ algorithm. Figure 13 illustrates the technique applied at a single time step.

Having sorted the particle cloud to obtain more accurate centroids, Eq. (15) determines the instantaneous angle between the estimated boresight of the SO, the $-Y$ facet of IS-18, and each hypothesized line-of-sight vector. Figure 14a illustrates which hypothesis had the smallest instantaneous angle during the observation window. The y axis labels each hypothesis; and the true subject of the target SO (16111) is at the top of the chart. Figure 14b illustrates the instantaneous angle for all five SOs that exhibited the smallest instantaneous angle for at least one time step. Here, the y axis is the instantaneous angle and not the index of the hypothesized subject. Inspection of both Figs. 14a and 14b reveals that, by using this instantaneous angle, the true subject is never identified, even when starting close to the right hypothesis.

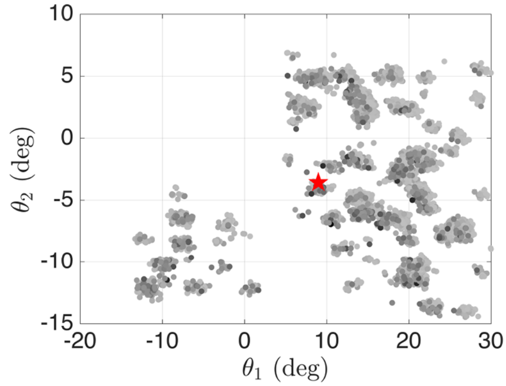
This fact motivated the MDM, given by Table 3. Inspection of the time-averaged MDM revealed that the true subject tracked by the target had the second smallest mean MDM. However, the error between the true subject and that which was predicted most likely by the MDM was 0.05 deg. There are several cases where both the instantaneous angle or the average MDM can give incorrect results. The first case is subjects for which the orbits pass serendipitously close to the true subject for arbitrary durations of time from the perspective of the target SO. This is the case for object 19120, as seen in Fig. 14a. Another case occurs when the SO happens to be in a very similar orbit plane as the true subject SO, perhaps only differentiating by a small offset in Euclidian space. This is the case for object 21397. Because of instances like these, the ability to estimate the body angular velocities is a crucial element of successful SO operational mode classification.

The first proposed metric that relies on this extra information is the Mahalanobis distance. Figure 15 plots the hypothesis with the smallest MD at each time step. Inspection of the figure reveals that the true subject (16111) is correctly identified by the smallest MD in 77% of the time steps. The other SO identified by a small MD (21397) is in an orbit very similar to the true subject. To further examine how similar the orbits of these two hypotheses appear to the target SO, the momentum analog is used.

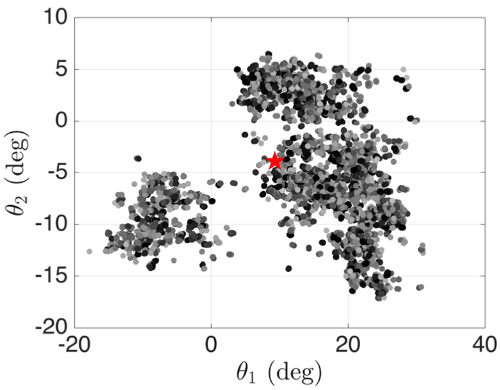
The two-sided hypothesis test for equal means is carried out at each time step of the observation. Figure 16 illustrates the t-statistic value of each hypothesis as dashed black lines, and the critical region demarcation as a solid black line at the $\alpha = 0.1$ significance level. Figure 17 plots the same information, where those hypotheses that have not been rejected as likely observation targets are plotted as a function of time.



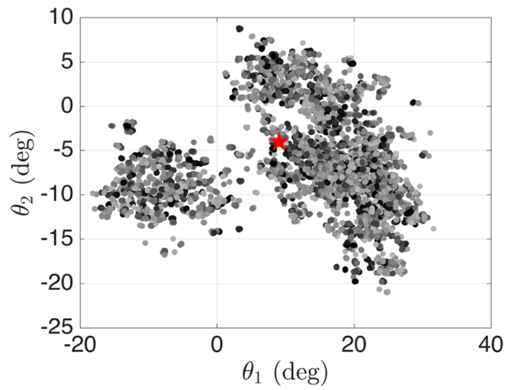
a) 5 October 2015 08:05:00 UTC



b) 5 October 2015 08:08:06 UTC

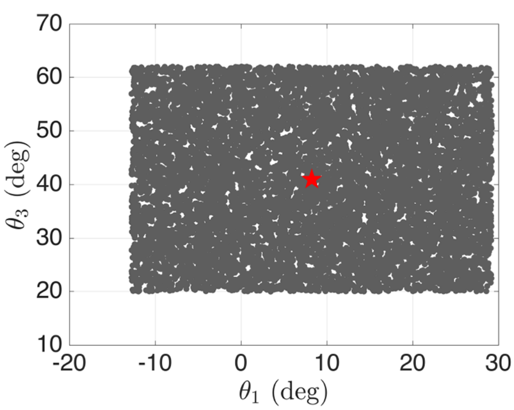


c) 5 October 2015 08:11:32 UTC

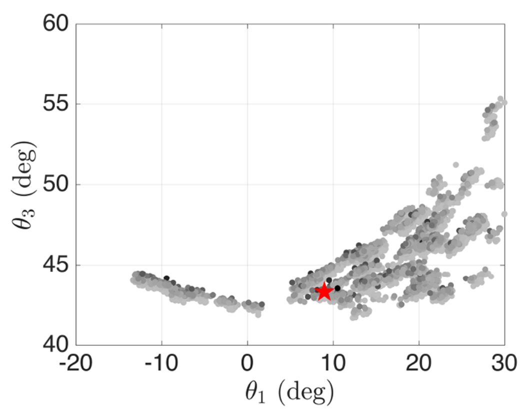


d) 5 October 2015 08:15:09 UTC

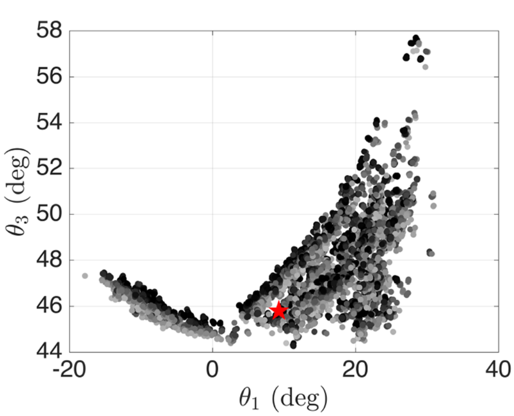
Fig. 9 2-D view of θ_1–θ_2.



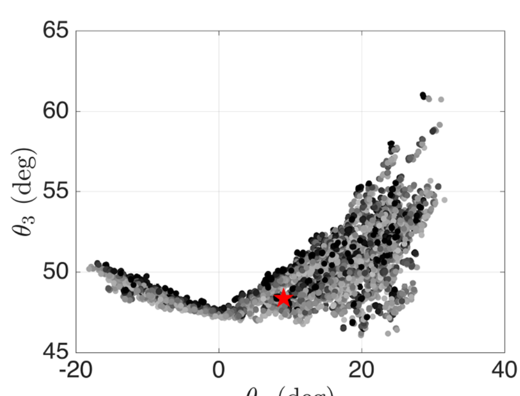
a) 5 October 2015 08:05:00 UTC



b) 5 October 2015 08:08:06 UTC



c) 5 October 2015 08:11:32 UTC



d) 5 October 2015 08:15:09 UTC

Fig. 10 2-D view of θ_1–θ_2.

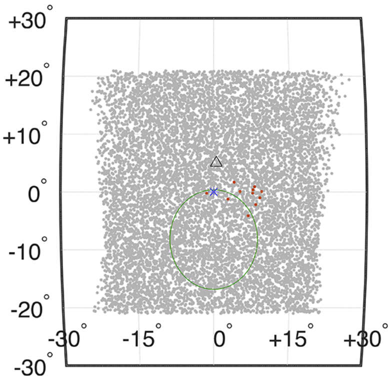


Fig. 11 5 October 2015, 08:05:00 UTC.

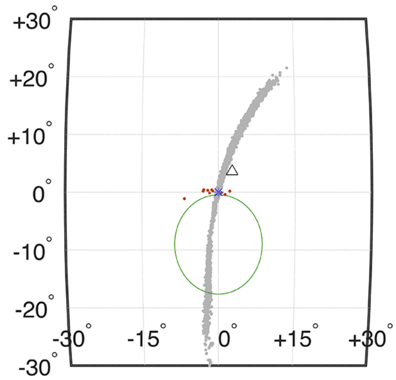


Fig. 12 5 October 2015, 08:15:09 UTC.

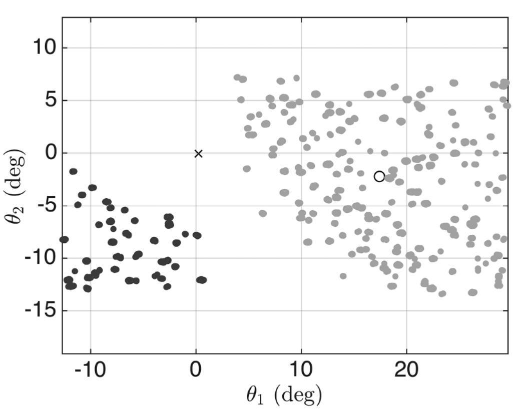


Fig. 13 2-D view of attitude states sorted via *k* means.

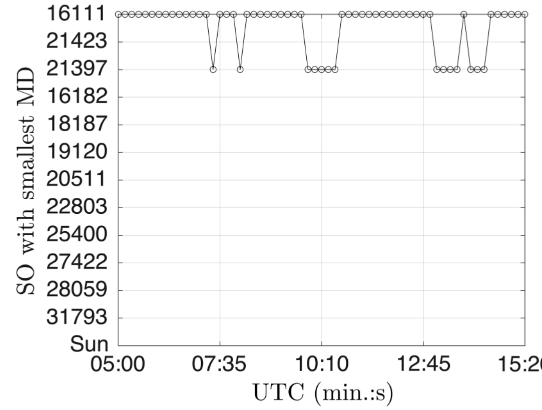


Fig. 15 SO with smallest Mahalanobis distance.

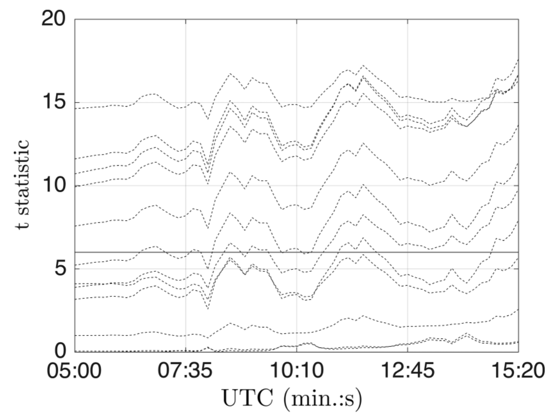


Fig. 16 Varying *t* statistic of each hypothesis.

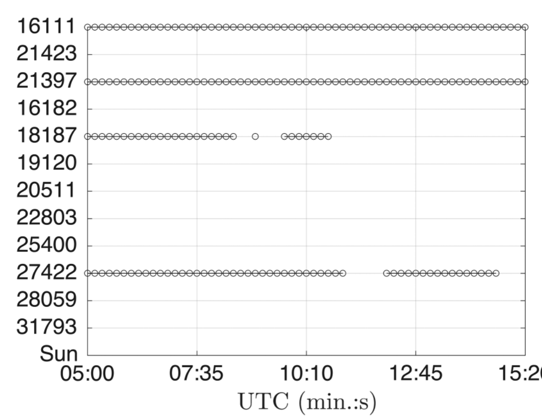
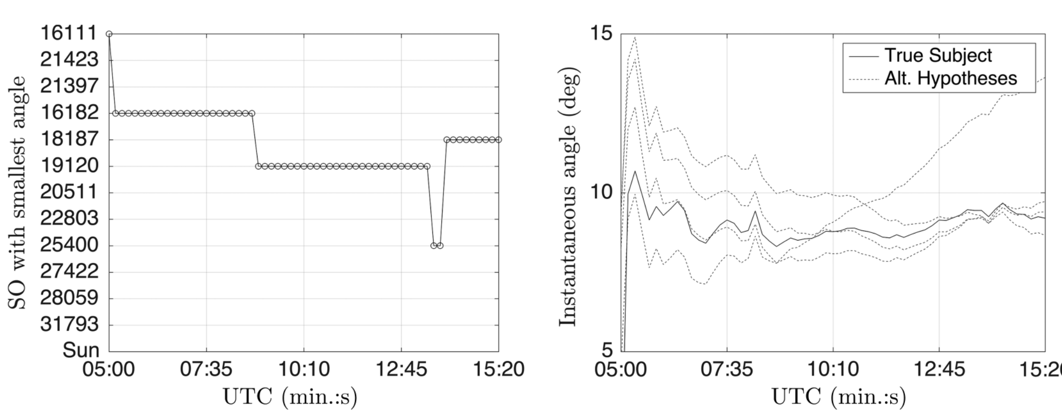


Fig. 17 Angular momentum classification.



a) Hypothesis with smallest instantaneous angle **b) Instantaneous angle for five selected hypothesis**

Fig. 14 Classifying active control mode with attitude states.

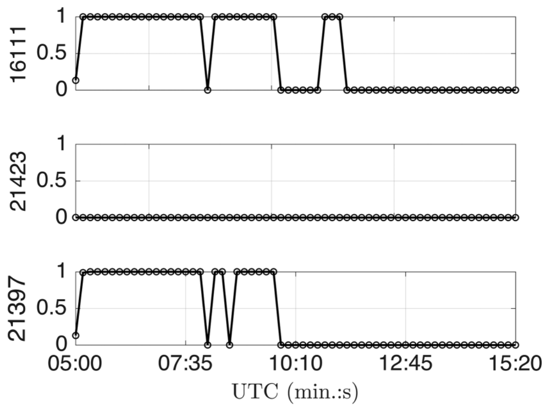


Fig. 18 Classification based on SNR.

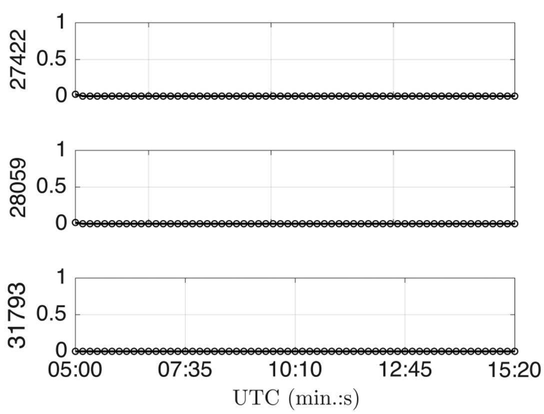


Fig. 19 Classification based on SNR.

Figure 17 shows that, given the estimated covariance of the body angular velocities, the mean body angular velocities required to track objects 16111 and 21397 are equivalent over the entire observation period. Two additional objects also periodically appear to have the same velocities. This is due to the fact that both of these objects are moving predominantly in the direction of the body y axis, which has been neglected from these calculations because it is unobservable. This methodology reinforces the general philosophy that observational evidence can only be used to eliminate hypotheses.

The results for the final proposed methodology are summarized in Figs. 18 and 19. These figures show the percentage of the angular velocity posterior distribution that is below the angular velocity limit of $\omega_e = 4 \cdot 10^{-5}$ rad/s, which result in SO subject streaks having SNR $\geq$15. In lieu of more detailed modeling, every object is assumed to have a $16m_v$ apparent visual magnitude with respect to the target SO. Table 4 enumerates the other parameters assumed in this work to simulate an optical environment similar to those on orbit. For brevity, only the first and last three hypotheses are depicted, as all others had near-zero probability.

Table 4 System parameters

Parameter	Units	Value
Aperture diameter of target SO	m	0.2
Focal ratio of target SO	— —	6
Pixel size of target SO sensor	m	5e-6
Irradiance of magnitude 0 source	photons/s/m^2	$5.6 \cdot 10^{10}$
Atmospheric seeing/ mount jitter	arcsecond	0.5
Sky brightness	visual magnitudes per square arcsecond (m_v/as^2)	30
CCD quantum efficiency	— —	0.6
CCD dark current	electrons per pixel per second (e/pixel/s)	0.5
Algorithm required SNR	— —	15
Atmospheric transmittance	— —	1
Optical transmittance	— —	0.9
Secondary transmittance	— —	1

Examining Fig. 18 reveals that the true subject SO (16111) had the most time steps where 100% of the particles were below the threshold. As expected, the next most likely hypothesis is again 21397. The remaining hypotheses all appeared as in Fig. 19, with none of the particles meeting the SNR requirements.

V. Conclusions

Several methodologies for directly identifying the observational target of a target space object are presented. The ability of the Rao–Blackwellized particle filter to provide estimates of the body angular velocities as well as attitude states proves to be a key enhancement over previous techniques. Critically, the proposed contributions enable the discrimination of targets that are colocated along the space-object boresight. The fusion of these estimates enables the contributions of this work to offer more precise descriptions of the active control mode. By providing the target that the observed space object is tracking, rather than the categorical classifications previously described in the literature, analysts can infer the mission purpose of a target space object, which is a key component of space domain awareness.

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